

Automated Blackout Window Shade

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Course: ECET 230 Prototyping fundamentals

Team



Ezra Lynn

Engineer

Electrical and Computer Engineering Technology Major at NJIT

[LinkedIn](#)

El364@njit.edu

EzraLynn.com



Qosai Awawda

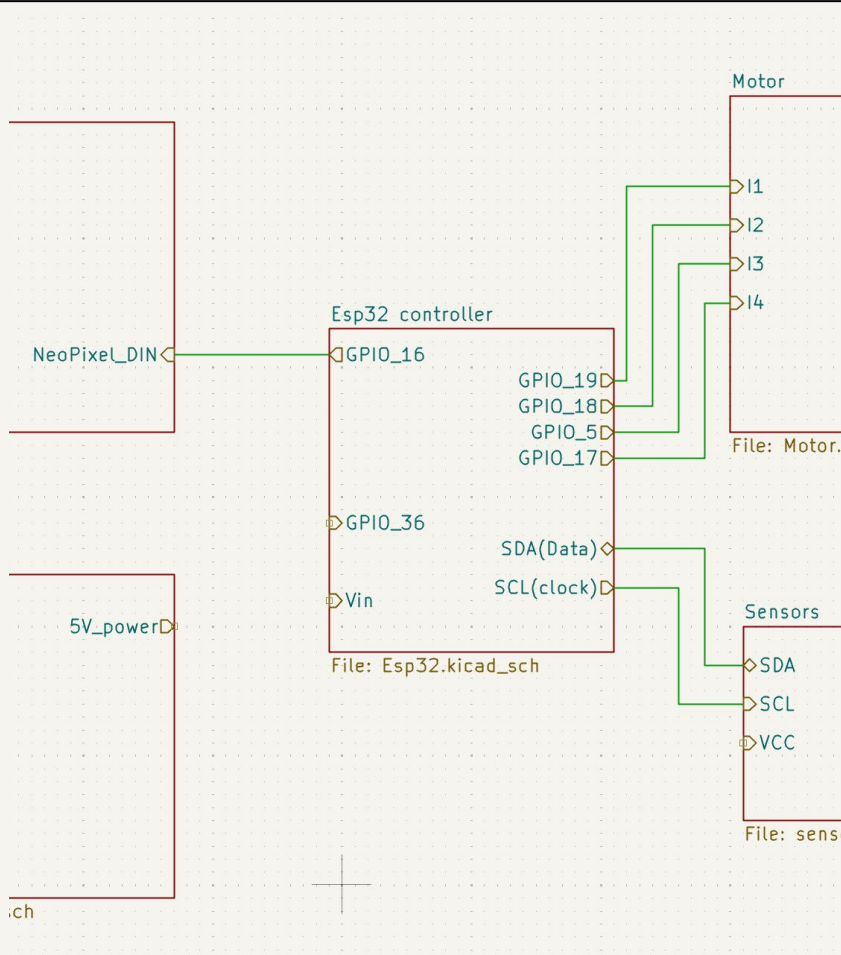
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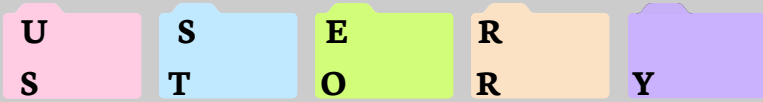
[LinkedIn](#)

qma2@njit.edu

AGENDA



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"As a student whose degree depends on a 6:50 AM wakeup,
I want my environment to automate my wake-up call when my alarm
doesn't do the job"

"As someone who forgets that windows work both ways,
I want my room to sense the darkness and seal itself shut,
So that my neighbors aren't treated to a front-row seat of my life in my
pajamas."

Objectives

Objective I

Our shade must rise and lower

- **Motorized**

Objective II

Our Shade must be controlled

- **Remotely**
- **Manually**
- Based on **light sensor**

Objective III

Our shade must be part of an **Automated** system to maximize efficiency

Objective IV

With outlets not usually close to windows, our shade must be

- **Battery powered**
- **Rechargable**

Component Selection

Automation

ESP32

- Integrated Bluetooth and wifi
- Compatible with all components
- Similar to arduino - within technical expertise

Motorized

28BYJ-48 - 5V STEPPER MOTOR

- Already own - easy and free to use for prototyping
- Enough Torque to raise and lower shade
- Used previously with arduino - within technical expertise

Controls

iPhone + Push Buttons + Light Sensors
BH1750

- Already own - easy and free to use for prototyping
- User must be able to open manually in addition to automated system
- Used previously with arduino - within technical expertise

Power

Battery pack + Solar panel FIT0333

- Battery pack for mounting location freedom
- Solar panel to recharge battery pack
- The batteries are rechargeable so that user can charge if solar was insufficient

Simple Workflow: less components = less room for error



Four Stage Rollout + Demonstration

	Stage 1	Stage 2	Stage 3	Stage 4
Arduino Uno	✓	✓	✗	✗
ESP32	✗	✗	✓	✓
Push Buttons	✓	✓	✓	✓
Light Sensor	✗	✗	✗	✓
Battery Pack	✗	✗	✗	✓
3D Gear System	✗	✓	✓	✓

Conclusion

MotoShadex - “A reliable, solar rechargeable, battery powered, motorized blackout shade to streamline morning routine and maximize efficiency, privacy, and alarm effectiveness.”

GitHub Repository:

https://github.com/travelsquad/ECET_230_class_project/tree/main

Goodnotes Logbook:

<https://web.goodnotes.com/s/00kup6R6PLASRnGxm8Rgop>